Pharma Data Engineering Platform

Analytics Engine for Pharma Supply Chain

Technical Design Document

1. Project Abstract

The Pharma Data Engineering Platform is a cloud-native analytics solution built on AWS that provides end-to-end visibility into the pharmaceutical supply chain. The platform ingests data from two complementary paths: a real-time streaming pipeline for operational events and a batch pipeline for master, transactional history and snapshot data.

All data is stored in Amazon S3 organised as a datalake with Parquet format. Queries are executed via AWS Athena, and business intelligence dashboards are delivered through Amazon QuickSight connected directly to Athena as its data source. This architecture eliminates the need for a dedicated data warehouse while still providing interactive query performance through Athena partition pruning and columnar storage.

The platform serves three primary stakeholders: Supply Chain Operations, Quality Assurance and Regulatory Affairs, and Commercial / Sales Analytics. Each stakeholder group accesses pre-built QuickSight dashboards that are refreshed continuously from the streaming layer and once daily from the batch layer.

2. Use Case and Business Intention

2.1 Business Problem

Pharmaceutical supply chains face unique compliance and operational challenges that generic supply chain platforms do not address:

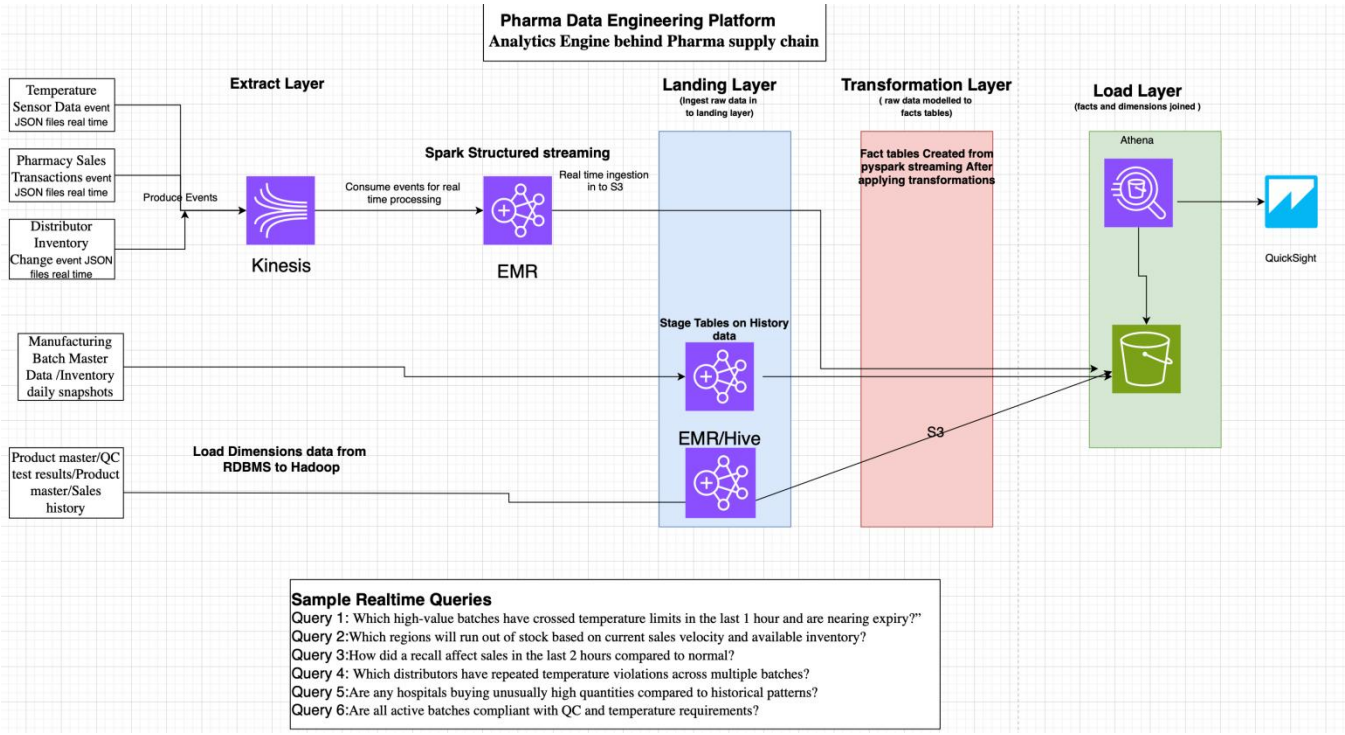
- Cold chain integrity must be maintained and evidenced for temperature-sensitive products such as vaccines, insulin, and biologics from point of manufacture to point of dispensing.
- Batch traceability is a regulatory requirement. Every unit sold must be traceable back to its manufactured batch, its QC test results, and its storage history.
- Distributor stock visibility is fragmented. Without a unified platform, stock-outs and near-expiry write-offs go undetected until they impact patients or revenue.
- QC failure impact assessment is manual and slow. When a batch fails post-release testing, identifying affected in-transit orders currently takes days.

2.2 Platform Intention

This platform addresses all four problems by:

- Streaming temperature sensor readings in real time and immediately comparing them against the cold chain specification of the batch being stored, alerting operations within seconds of a breach.
- Building a unified batch_id spine that links every temperature reading, every sale, every QC result, and every inventory movement to the same manufactured lot.
- Computing inventory coverage (days of stock) continuously by joining real-time stock movement events with historical sales velocity, enabling proactive replenishment.
- Cross-referencing QC failure records against live order fulfilment status to instantly produce a recall impact list with revenue at risk.

3. Data Flow Architecture



3.1 Overview

Data flows through four logical layers before reaching QuickSight dashboards. The technology choices at each layer are listed below.

Layer	Technology	Input	Output
Extract	Kinesis Data Streams (streaming) / JDBC + S3 copy (batch)	IoT sensors, POS systems, WMS, ERP	Raw JSON / Parquet events on Kinesis or S3
Landing	EMR Spark Structured Streaming / EMR Hive	Kinesis stream, S3 raw files	Cleansed Parquet on S3 (Delta Lake)
Transformation	Athena CTAS	S3 landing zone	Fact and dimension Parquet tables on S3
Load and Visualise	Athena + QuickSight	S3 fact and dimension tables	Interactive dashboards and reports

3.2 Streaming Data Flow

Three event types are produced by operational systems and streamed through Kinesis:

- Temperature sensor events are emitted every 30 seconds per device. Spark Structured Streaming consumes these events, normalises temperature units to Celsius, computes a 5-minute rolling window average, flags anomalies beyond 3 standard deviations, and marks breaches against the batch cold chain specification before writing enriched Parquet to S3.

- Pharmacy sales transactions are produced on every point-of-sale event. Spark deduplicates on transaction ID using a 10-minute watermark, enriches each record with product category and controlled-substance flag via a broadcast join against the product dimension, and writes to the sales fact table on S3.
- Distributor inventory change events are produced on every stock movement. Spark computes days-of-stock coverage, flags records where current stock falls below the reorder point, and routes those flagged records to an SNS alert topic in addition to writing to S3.

3.3 Batch Data Flow

Five batch sources are loaded daily via JDBC export to S3 followed by EMR/Hive processing:

- Manufacturing batch master: full extract from ERP at 02:00 IST. Contains cold chain specifications, batch quantities, and release status for every manufactured lot.
- Inventory daily snapshot: closing stock balances per distributor per product at 03:00 IST.
- Product master: full refresh of product catalogue, drug schedules, controlled substance flags, and pricing at 01:00 IST.
- QC test results: incremental append of laboratory results
- Sales history: historical transaction records pre-dating the streaming platform, used for trend analysis and 7-day sales velocity calculations.

3.4 S3 Storage Layout

All data is partitioned by date for efficient Athena partition pruning:

- s3://pharma-datalake/landing/ — raw events and daily extracts
- s3://pharma-datalake/facts/ — cleansed streaming and batch fact tables
- s3://pharma-datalake/dimensions/ — dimension and master data tables
- s3://pharma-datalake/alerts/ — cold chain and stock alert records

4. Table Schemas

4.1 Streaming: Temperature Sensor Events

Source: IoT cold chain sensors. Transport: Kinesis. Frequency: Every 30 seconds per device.

Field	Type	Nullable	Description
event_id	String	No	Unique event identifier — primary deduplication key
event_time	Timestamp	No	UTC reading timestamp from the sensor device
sensor_id	String	No	Physical sensor identifier
batch_id	String	No	Batch being monitored — FK to dim_batch_master
location	String	Yes	Warehouse or vehicle zone label
temperature_c	Decimal	No	Temperature in Celsius (normalised by Spark)
humidity_pct	Decimal	Yes	Relative humidity percentage
unit	String	No	Original unit: celsius or fahrenheit
alert_flag	Boolean	No	Set by Spark if temperature outside batch range
anomaly_flag	Boolean	No	Set by Spark if reading exceeds 3-sigma window threshold
window_avg_c	Decimal	Yes	5-minute rolling window average (Spark computed)
partition_date	Date	No	Partition column for S3 and Athena pruning

4.2 Streaming: Pharmacy Sales Transactions

Source: Pharmacy POS systems. Transport: Kinesis. Frequency: Real-time on every sale.

Field	Type	Nullable	Description
transaction_id	String	No	POS transaction identifier — deduplication key
event_time	Timestamp	No	UTC sale timestamp
pharmacy_id	String	No	Pharmacy outlet identifier — FK to dim_pharmacy
product_sku	String	No	Product sold — FK to dim_product
batch_id	String	No	Batch dispensed — FK to dim_batch_master
prescription_no	String	Yes	Rx number — null for over-the-counter products
quantity_sold	Integer	No	Number of units sold
unit_price_inr	Decimal	No	Selling price per unit in INR
discount_pct	Decimal	Yes	Discount percentage applied (default 0)
revenue_inr	Decimal	No	Net revenue after discount
controlled_flag	Boolean	No	Enriched by Spark from dim_product broadcast join
product_category	String	Yes	Enriched by Spark from dim_product broadcast join
fulfillment_status	String	No	PENDING / IN_TRANSIT / COMPLETED
partition_date	Date	No	Partition column for S3 and Athena pruning

4.3 Streaming: Distributor Inventory Change Events

Source: Distributor WMS systems. Transport: Kinesis. Frequency: Real-time on every stock movement.

Field	Type	Nullable	Description
event_id	String	No	Unique movement event identifier
event_time	Timestamp	No	UTC movement timestamp
distributor_id	String	No	Distributor identifier — FK to dim_distributor
product_sku	String	No	Product moved — FK to dim_product
batch_id	String	Yes	Batch reference if tracked at this level
change_type	String	No	RECEIPT / DISPATCH / ADJUSTMENT / RETURN
quantity_delta	Integer	No	Positive for stock in, negative for stock out
current_stock	Integer	No	Stock level after this movement
reorder_point	Integer	Yes	Minimum reorder threshold for this product
below_reorder	Boolean	No	Computed by Spark: current_stock less than reorder_point
days_cover	Decimal	Yes	Computed by Spark: stock divided by 7-day avg daily sales
distributor_region	String	Yes	Enriched by Spark from dim_distributor broadcast join
partition_date	Date	No	Partition column for S3 and Athena pruning

4.4 Batch: Manufacturing Batch Master

Source: Manufacturing ERP. Transport: JDBC to S3, processed by EMR/Hive. Frequency: Daily full extract at 02:00 IST.

Field	Type	Nullable	Description
batch_id	String	No	Unique batch identifier — natural key for all joins
product_sku	String	No	Product in this batch — FK to dim_product
plant_id	String	No	Manufacturing plant — FK to dim_plant
manufacture_date	Date	No	Date production was completed
expiry_date	Date	No	Batch expiry date
batch_qty	Integer	No	Total units produced
min_temp_c	Decimal	Yes	Cold chain minimum storage temperature allowed
max_temp_c	Decimal	Yes	Cold chain maximum storage temperature allowed
batch_status	String	No	RELEASED / QUARANTINE / RECALLED / EXPIRED
release_date	Date	Yes	QA release date — null if not yet released
recall_date	Date	Yes	Date recalled — null if not recalled
recall_reason	String	Yes	Description of recall reason

4.5 Batch: Daily Inventory Snapshot

Source: Distributor ERP / WMS. Transport: JDBC to S3. Frequency: Daily closing balances at 03:00 IST.

Field	Type	Nullable	Description
snapshot_date	Date	No	Date of this closing balance snapshot
distributor_id	String	No	Distributor — FK to dim_distributor
product_sku	String	No	Product — FK to dim_product
batch_id	String	Yes	Batch tracked at this level if available
opening_stock	Integer	No	Stock at start of day
receipts	Integer	No	Units received during the day
dispatches	Integer	No	Units dispatched during the day
adjustments	Integer	No	Write-offs and corrections
returns	Integer	No	Units returned from pharmacies
closing_stock	Integer	No	Calculated: opening plus receipts minus dispatches plus returns plus adjustments
reorder_point	Integer	Yes	Minimum reorder threshold

4.6 Dimension: Product Master

Source: Product MDM system. Frequency: Daily full refresh at 01:00 IST.

Field	Type	Nullable	Description
product_sku	String	No	Unique product code — natural key
product_name	String	No	Full commercial name
generic_name	String	Yes	INN or generic name
category	String	Yes	Antibiotic / Vaccine / Analgesic / Cardiac etc.
dosage_form	String	Yes	Tablet / Capsule / Injection / Syrup
strength	String	Yes	e.g. 500mg, 1g per 5ml
schedule	String	Yes	India drug schedule: H / H1 / G / X
controlled_flag	Boolean	No	True for narcotic or psychotropic substances
cold_chain_required	Boolean	No	True if 2 to 8 degree Celsius storage required
min_temp_c	Decimal	Yes	Minimum storage temperature (cold chain products)
max_temp_c	Decimal	Yes	Maximum storage temperature (cold chain products)
reorder_point_units	Integer	Yes	Default reorder threshold in units
unit_cost_inr	Decimal	Yes	Manufacturer cost price in INR
unit_price_mrp_inr	Decimal	Yes	Maximum retail price in INR
is_active	Boolean	No	False for discontinued products

4.7 Dimension: QC Test Results

Source: Laboratory Information Management System (LIMS). Frequency: Daily incremental append.

Field	Type	Nullable	Description
qc_id	String	No	Unique QC test record identifier
batch_id	String	No	Batch tested — FK to dim_batch_master
product_sku	String	No	Product — FK to dim_product
test_date	Date	No	Date test was conducted
lab_id	String	Yes	Laboratory that ran the test
potency_pct	Decimal	Yes	Percentage of labelled strength found
dissolution_pct	Decimal	Yes	Dissolution test result percentage
sterility_result	String	Yes	PASS / FAIL / NA
endotoxin_result	String	Yes	PASS / FAIL / NA
overall_result	String	No	PASS or FAIL
failure_reason	String	Yes	Description of failure — null if PASS
retested_flag	Boolean	No	True if this record is a retest
retest_result	String	Yes	PASS or FAIL result of the retest
approved_by	String	Yes	QA officer who approved the release

5. Data Cleansing and Validation Rules

5.1 Streaming Table Rules

Temperature Sensor Events

Rule	Field	Condition	Action on Failure
Not null check	event_id, event_time, sensor_id, batch_id, temperature_c	Any of these fields is null or empty	Drop record, write to dead-letter S3 path with error code NULL_KEY
Unit normalisation	unit, temperature_c	Unit equals fahrenheit	Convert: celsius = (fahrenheit minus 32) multiplied by 5 divided by 9. Set unit to celsius
Temperature range guard	temperature_c	Value is below minus 100 or above 200 degrees Celsius	Drop as sensor error, write to dead-letter path with error code SENSOR_RANGE
Batch existence check	batch_id	batch_id not found in broadcast of dim_batch_master	Flag record as UNKNOWN_BATCH, still write but exclude from compliance reports
Cold chain breach flag	temperature_c, batch min/max	temperature_c outside batch min_temp_c to max_temp_c range	Set alert_flag = true, emit to SNS cold chain alert topic
Humidity range guard	humidity_pct	Value outside 0 to 100 percent	Set humidity_pct to null, log warning
Event time skew	event_time	Event time is more than 60 minutes behind current processing time	Tag as LATE_ARRIVAL, still process but flag for audit

Pharmacy Sales Transactions

Rule	Field	Condition	Action on Failure
Not null check	transaction_id, event_time, pharmacy_id, product_sku, quantity_sold, revenue_inr	Any of these fields is null	Drop record, write to dead-letter path with error code NULL_KEY
Quantity validation	quantity_sold	Value is zero or negative	Drop record with error code INVALID_QUANTITY
Revenue validation	revenue_inr, quantity_sold, unit_price_inr	revenue_inr differs from quantity times unit_price times (1 minus discount) by more than 1 INR	Recompute revenue_inr from components and log discrepancy
Price range check	unit_price_inr	Price is zero or negative or greater than 100,000 INR	Set to null and flag for manual review
Discount range check	discount_pct	Value outside 0 to 100	Clamp to valid range, log warning
Product	product_sku	SKU not found in broadcast of	Set controlled_flag to false and

Rule	Field	Condition	Action on Failure
enrichment		dim_product	product_category to UNKNOWN, still write
Fulfillment status check	fulfillment_status	Value not in allowed list PENDING, IN_TRANSIT, COMPLETED	Set to COMPLETED as default, log warning
Future event guard	event_time	event_time is more than 5 minutes in the future	Adjust to current processing time and log TIME_SKEW

Distributor Inventory Change Events

Rule	Field	Condition	Action on Failure
Not null check	event_id, event_time, distributor_id, product_sku, change_type, quantity_delta, current_stock	Any of these fields is null	Drop record,
Change type validation	change_type	Value not in RECEIPT, DISPATCH, ADJUSTMENT, RETURN	Drop record with error code INVALID_CHANGE_TYPE
Delta sign validation	change_type, quantity_delta	RECEIPT or RETURN has negative delta, or DISPATCH has positive delta	Flip sign, log warning SIGN_MISMATCH
Stock floor check	current_stock	Value is negative	Set to 0 and log NEGATIVE_STOCK warning — physically impossible
Days cover computation	current_stock, reorder_point	Reorder point is present	Compute days_cover from Spark state store using rolling 7-day avg sales
Below reorder flag	current_stock, reorder_point	current_stock less than reorder_point	Set below_reorder = true, emit stock alert to alert dir
Distributor enrichment	distributor_id	Distributor not found in dim_distributor broadcast	Still write with region set to UNKNOWN, exclude from regional reports
Duplicate suppression	event_id	Same event_id within 10-minute window	Drop duplicate

5.2 Batch Table Rules

Manufacturing Batch Master

Rule	Field	Condition	Action on Failure
Primary key uniqueness	batch_id	Duplicate batch_id in daily extract	Keep latest record by load timestamp, log DUP_BATCH_ID
Not null check	batch_id, product_sku, plant_id, manufacture_date, expiry_date, batch_qty	Any of these is null	Reject record, move to quarantine S3 path
Date logic check	manufacture_date, expiry_date	expiry_date is on or before manufacture_date	Reject record with error code INVALID_DATE_RANGE
Quantity floor	batch_qty	batch_qty is zero or negative	Reject record with INVALID_QUANTITY
Cold chain consistency	cold_chain_required, min_temp_c, max_temp_c	dim_product shows cold_chain_required = true but min_temp_c or max_temp_c is null in batch	Default to product-level temperature range and log TEMP_RANGE_DEFAULTED
Temperature range logic	min_temp_c, max_temp_c	Both present but min_temp_c is greater than or equal to max_temp_c	Reject record with INVALID_TEMP_RANGE
Status validation	batch_status	Value not in RELEASED, QUARANTINE, RECALLED, EXPIRED	Default to QUARANTINE and log UNKNOWN_STATUS
Expiry proximity alert	expiry_date	expiry_date within 90 days of load date	Tag with near_expiry_flag = true for reporting

Daily Inventory Snapshot

Rule	Field	Condition	Action on Failure
Not null check	snapshot_date, distributor_id, product_sku, closing_stock	Any of these is null	Reject record, move to quarantine S3 path
Balance equation check	opening_stock, receipts, dispatches, adjustments, returns, closing_stock	closing_stock not equal to opening + receipts minus dispatches + adjustments + returns	Log BALANCE_MISMATCH and recompute closing_stock from components
Stock floor check	closing_stock, opening_stock	Any stock value is negative	Set to 0 and log NEGATIVE_STOCK
Distributor existence	distributor_id	Not in dim_distributor	Write with flag UNKNOWN_DISTRIBUTOR, exclude from regional aggregations
Product existence	product_sku	Not in dim_product	Write with flag UNKNOWN_PRODUCT,

Rule	Field	Condition	Action on Failure
			exclude from product reports
Duplicate date key	snapshot_date + distributor_id + product_sku	Same composite key appears twice	Keep record with higher closing_stock, log DUP_SNAPSHOT
Stale snapshot guard	snapshot_date	snapshot_date is more than 3 days before load date	Log STALE_SNAPSHOT and alert data engineering team

Product Master

Rule	Field	Condition	Action on Failure
Primary key uniqueness	product_sku	Duplicate SKU in daily extract	Keep latest by valid_from date, log DUPLICATE_SKU
Not null check	product_sku, product_name, controlled_flag, cold_chain_required, is_active	Any of these is null	Reject record
Cold chain completeness	cold_chain_required, min_temp_c, max_temp_c	cold_chain_required = true but min_temp_c or max_temp_c is null	Flag INCOMPLETE_COLD_CHAIN, default to 2–8 degrees Celsius range
Temperature logic	min_temp_c, max_temp_c	Both present and min greater than or equal to max	Reject with INVALID_TEMP_RANGE
Price positivity	unit_cost_inr, unit_price_mrp_inr	Either price is zero or negative when not null	Set to null and log INVALID_PRICE
Schedule validation	schedule	Value not in H, H1, G, X, OTC or null	Set to null, log UNKNOWN_SCHEDULE

QC Test Results

Rule	Field	Condition	Action on Failure
Not null check	qc_id, batch_id, product_sku, test_date, overall_result	Any of these is null	Reject record
Overall result validation	overall_result	Value not PASS or FAIL	Reject with INVALID_QC_RESULT
Failure reason consistency	overall_result, failure_reason	overall_result = FAIL but failure_reason is null	Flag MISSING_FAILURE_REASON for QA review
Potency range check	potency_pct	Value outside 0 to 120 percent when not null	Flag POTENCY_OUT_OF_RANGE for QA review
Batch existence check	batch_id	batch_id not in dim_batch_master	Write with flag ORPHAN_QC_RESULT, exclude from compliance joins

Rule	Field	Condition	Action on Failure
Test date logic	test_date	test_date is before batch manufacture_date or after expiry_date	Reject with INVALID_TEST_DATE
Retest consistency	retested_flag, retest_result	retested_flag = true but retest_result is null	Flag INCOMPLETE_RETEST for QA review

6. Business Insight Queries (Athena)

The following queries are described in business terms. All queries execute in Athena against Parquet tables on S3. QuickSight SPICE datasets are built from these queries.

6.1 Cold Chain Compliance — Real-Time Breach Detection

Purpose: Identify which batches have had temperature readings outside their allowed cold chain range in the last 24 hours. Supports immediate operational response and regulatory event reporting.

Join: fact_temperature_events joined to dim_batch_master on batch_id.

- Filter to events in the last 24 hours using partition_date and event_time.
- Bring in the batch's allowed minimum and maximum temperature from dim_batch_master.
- Classify each reading as OK or BREACH by comparing temperature_c against the batch's min and max range.
- Return sensor ID, location, actual reading, allowed range, batch and product details, and breach status.
- Order by event time descending so most recent breaches appear first.

6.2 Stock Coverage by Distributor

Purpose: Show how many days of stock each distributor holds for each product, based on recent sales velocity. Enables proactive replenishment before stock-outs occur.

- Join fact_inventory_snapshot (batch) to fact_sales_transactions (streaming) on product_sku and distributor_id.
- Sum quantity_sold over the last 7 days to get recent sales volume.
- Compute days of cover as closing_stock divided by the 7-day average daily sales rate.
- Join dim_distributor for region and tier, and dim_product for product name and category.
- Group by product, distributor, region, and tier.
- Flag records where days_cover is below 7 as critical, between 7 and 14 as low, and above 14 as adequate.

6.3 QC Failure Impact on Live Orders

Purpose: When a batch fails QC testing, instantly surface all sales transactions for that batch that are still in transit or pending fulfilment. Provides a rapid recall impact assessment with revenue at risk.

- Join dim_qc_results to fact_sales_transactions on batch_id.
- Filter QC records to overall_result = FAIL only.
- Filter sales records to fulfillment_status IN PENDING or IN_TRANSIT only.
- Bring in dim_product for product name and controlled flag.
- Count affected transactions and sum revenue_inr per failed batch.
- Order by revenue at risk descending.

6.4 Real-Time Sales Velocity by Product and Region

Purpose: Monitor how quickly each product is selling across regions in real time. Supports demand sensing and dynamic replenishment decisions by commercial teams.

- Query fact_sales_transactions filtered to the last 7 days using partition_date.
- Join dim_product for category and product name.
- Join dim_pharmacy for city, state, and zone.
- Aggregate total quantity sold and total revenue by product, zone, and day.
- Compute 7-day rolling total and day-over-day percentage change for each product-region combination.
- Surface the top 10 fastest-growing and top 10 fastest-declining product-region pairs.

6.5 Batch Traceability — End-to-End Lot History

Purpose: Given a single batch ID, retrieve the complete chain of custody: manufacture details, QC test results, cold chain readings, distributor inventory movements, and pharmacy sales. Required for regulatory audit and recall support.

- Accept batch_id as a parameter.
- Retrieve manufacture, expiry, plant, and release details from dim_batch_master.
- Retrieve all QC test results for this batch from dim_qc_results.
- Retrieve all temperature readings for this batch from fact_temperature_events, including breach and anomaly flags.
- Retrieve all inventory movements for this batch from fact_inventory_events.
- Retrieve all pharmacy sales for this batch from fact_sales_transactions, including fulfilment status.
- Present results as separate sections: manufacture summary, QC summary, cold chain summary, distribution movements, and sales summary.

6.6 Near-Expiry Inventory Risk

Purpose: Identify stock that will expire within 90 days and cross-reference with sales velocity to estimate write-off risk. Enables targeted promotions or redistribution before expiry.

- Join dim_batch_master to fact_inventory_snapshot on batch_id.
- Filter to batches where expiry_date is within 90 days of today.
- Join fact_sales_transactions to compute the 30-day sales rate for each product per distributor.
- Estimate projected remaining stock at expiry date using current_stock minus projected sales.
- Flag as high risk where projected remaining stock at expiry is greater than 20 percent of current stock.
- Return product, distributor, expiry date, current stock, 30-day sales rate, and estimated write-off quantity.

7. QuickSight Reports and Dashboards

QuickSight is connected to Athena as its primary data source. SPICE datasets are used for high-volume historical analyses. Direct Athena queries are typically used for real-time operational dashboards where data freshness is critical.

7.1 Cold Chain Monitoring Dashboard

Audience: Supply chain operations, warehouse managers.

A. Breach timeline showing temperature readings against the allowed band for any selected batch over the last 24 hours.(Line chart with a reference band. X axis is event_time, Y axis is temperature_c. A horizontal reference band shaded in light red covers the area outside the batch's min_temp_c to max_temp_c range. The line shows actual readings. Breaches are visually obvious where the line exits the band.)

B. Alert summary table listing active breaches with sensor ID, location, deviation in degrees Celsius, duration in minutes, and batch details.(Table visual. Columns: sensor ID, location, current temperature, allowed range, deviation in degrees, breach duration in minutes, batch ID, product name. Conditional formatting turns the deviation cell red if breach is active. Sort by breach duration descending so the longest-running breach is always at the top.)

7.2 Distributor Stock and Replenishment Dashboard

Audience: Supply chain planners, distributor managers.

A. Stock coverage table ranked by days-of-cover ascending, filterable by product, region, and tier. Trend chart showing closing stock versus reorder point over the last 30 days for a selected product-distributor pair.(Dual-line chart. X axis is snapshot date for the last 30 days. Two lines per selected product-distributor pair: one for closing_stock and one for reorder_point. The reorder_point line is a flat reference line. When the closing_stock line dips below the reorder_point line the risk is visually clear.)

7.3 Sales Performance Dashboard

Audience: Commercial team, regional sales managers.

A. **Revenue by category donut chart** Donut chart. Each segment is one product category. Labels show category name and percentage of total. Centre label shows overall total revenue for the selected period. Simple, scannable, and appropriate for a five-category breakdown.

8. Technical Notes

8.1 Athena Configuration

- All fact tables are partitioned by partition_date. Athena queries must always include a partition_date predicate to avoid full-table scans.
- Glue Data Catalog is used as the Athena metadata store. All S3 tables are registered as Glue external tables.

Document prepared by the Data Engineering team. Classification: Confidential.

9. Definitions

What is cold chain compliance?

Cold chain compliance means maintaining a product within its required temperature range continuously from the point of manufacture to the point of dispensing to the patient — without any break in that controlled temperature environment.

The word "chain" is literal — it refers to an unbroken chain of controlled temperature custody. Every link in that chain matters. If any single link breaks — a warehouse zone gets too warm, a delivery vehicle's refrigeration unit fails, a pharmacist stores a vaccine in the wrong section of the fridge — the entire chain is considered compromised for that batch, regardless of how perfectly every other link was maintained.

Why it matters in pharma specifically

Most pharmaceutical products are chemically stable at room temperature. Cold chain applies specifically to products where the active molecule, the biological structure, or the physical formulation degrades outside a defined temperature window. The most common window in pharma is 2°C to 8°C — standard refrigerator temperature.

Products that require this include vaccines (where the antigen proteins denature above 8°C, rendering the vaccine ineffective without any visible change — it still looks like a normal liquid), insulin (where potency degrades above 8°C), certain biologics and monoclonal antibodies used in cancer treatment, and some blood products and diagnostic reagents.

The critical problem is that a cold chain failure is often invisible. A vial of vaccine that was exposed to 12°C for six hours looks exactly the same as one that was stored correctly the entire time. The patient receives it, it does not work, and depending on the disease that could mean anything from a preventable infection to a fatality. There is no warning label that appears on the product. This is why monitoring and compliance recording matters — the only way to know the product is still effective is to prove the temperature was maintained throughout.